

Isaac ROS Image distortion processing

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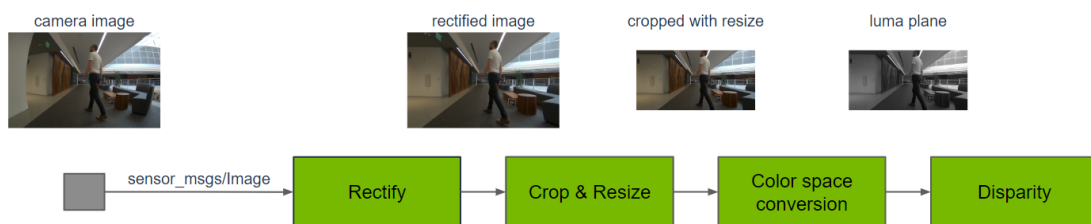
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Isaac ROS image distortion processing: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_image_pipeline/index.html

Overview

Isaac ROS image distortion processing uses the Isaac ROS image pipeline. Isaac ROS Image Pipeline is a metapackage of functionality for image processing. Camera output often needs pre-processing to meet the input requirements of multiple different perception functions. This can include cropping, resizing, mirroring, correcting for lens distortion, and color space conversion. For stereo cameras, additional processing is required to produce disparity between left + right images and a point cloud for depth perception.



Quick Experience

To simplify development, we primarily use the Isaac ROS Dev Docker image and demonstrate the effects there. This demonstration does not require any camera device installation; it simulates the camera data stream by playing a rosbag file.

Note: If you wish to install on your own device or connect a camera to develop other features, please refer to the Isaac ROS official website and connect a designated NVIDIA camera model for your own development.

Resize:

Open a terminal and enter the working directory.

```
cd ${ISAAC_ROS_WS}/src
```

Enter the Isaac ROS Dev Docker container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following startup command.

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py
••launch_fragments:=resize
```

Open a second terminal and enter the container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following command.

```
ros2 bag play --loop
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quickstart --remap
/hawk_0_left_rgb_image:=/image_raw /hawk_0_left_rgb_camera_info:=/camera_info
```

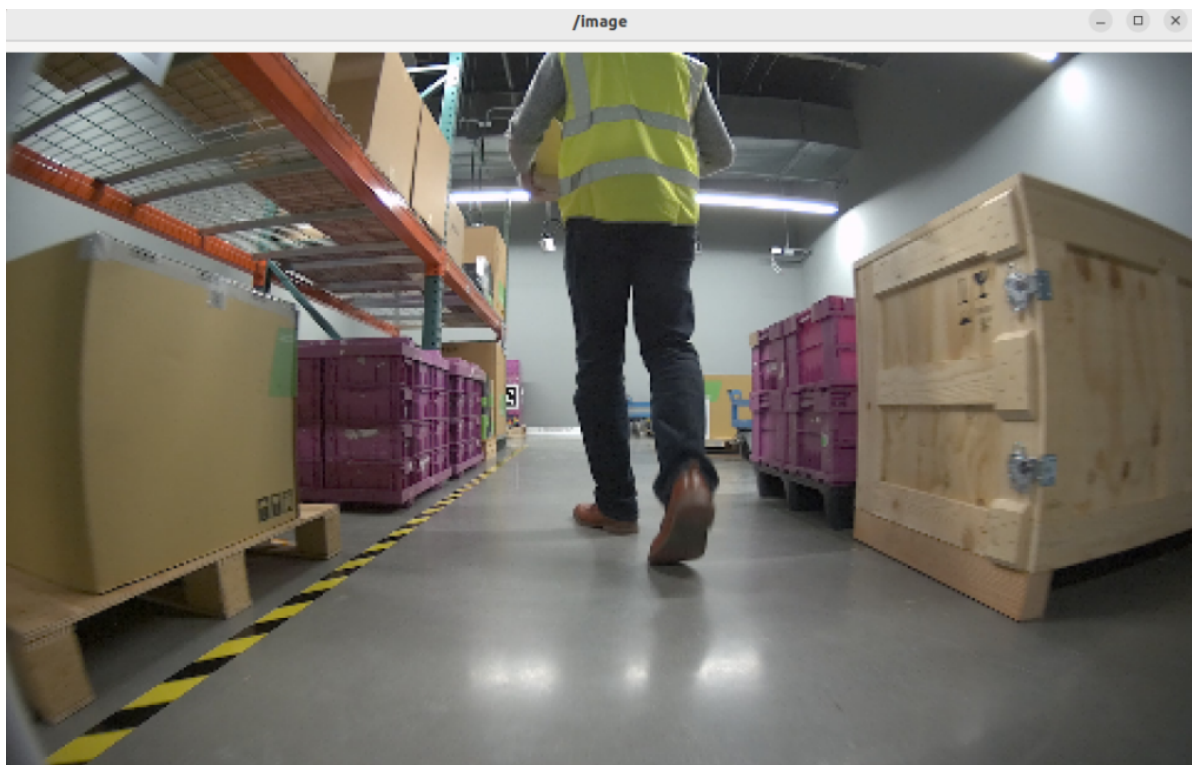
View the results

Open a third terminal and enter the container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following command:

```
ros2 run image_view image_view --ros-args --remap image:=resize/image
```



Color Conversion:

Open a terminal and enter the working directory.

Note: If you have already opened a container and run other commands, exit all Docker containers by typing `exit` in the first terminal before running the command.

```
cd ${ISAAC_ROS_WS}/src
```

Enter the Isaac ROS Dev Docker container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following startup command

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py  
••launch_fragments:=color_conversion  
interface_specs_file:=${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quic  
kstart_interface_specs.json
```

Open a second terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 bag play --loop  
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quickstart --remap  
/hawk_0_left_rgb_image:=/image_raw /hawk_0_left_rgb_camera_info:=/camera_info
```

View the run results

Open a third terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 run image_view image_view --ros-args --remap image:=image_mono
```



Crop:

Open a terminal and enter the working directory

```
cd ${ISAAC_ROS_WS}/src
```

Enter the Isaac ROS Dev Docker container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following startup command

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py  
••launch_fragments:=crop  
interface_specs_file:=${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quic  
kstart_interface_specs.json
```

Open a second terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 bag play --loop  
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quickstart --remap  
/hawk_0_left_rgb_image:=/image_raw /hawk_0_left_rgb_camera_info:=/camera_info
```

View the run results

Open a third terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 run image_view image_view --ros-args --remap image:=crop/image
```



Rectify:

Open a terminal and enter the working directory

```
cd ${ISAAC_ROS_WS}/src
```

Enter Isaac ROS Dev Docker container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following startup command

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py
••launch_fragments:=rectify_mono
interface_specs_file:=${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quic
kstart_interface_specs.json
```

Open a second terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following command

```
ros2 bag play --loop
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quickstart --remap
/hawk_0_left_rgb_image:=/image_raw /hawk_0_left_rgb_camera_info:=/camera_info
```

View the Run Results

Open a third terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following command

```
ros2 run image_view image_view --ros-args --remap image:=image_rect
```



Flip:

Open a terminal and enter the working directory

```
cd ${ISAAC_ROS_WS}/src
```

Enter Isaac ROS Dev Docker container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following startup command

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py
••launch_fragments:=flip
```

Open a second terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \
./scripts/run_dev.sh
```

Run the following command

```
ros2 bag play --loop  
${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_image_proc/quickstart --remap  
/hawk_0_left_rgb_image:=/image_raw /hawk_0_left_rgb_camera_info:=/camera_info
```

View the run results

Open a third terminal and enter the container

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command

```
ros2 run image_view image_view --ros-args --remap image:=image_flipped
```

